

3(a)	acceleration in opposite <u>direction</u> to displacement shown by – sign	B1
	g / L is constant	M1
	(so) acceleration is (directly) proportional to displacement	A1
3(b)	$\omega^2 = g / L$	C1
	$\omega = 2\pi / T$ or $\omega = 2\pi f$ and $f = 1 / T$	C1
	$(2\pi / T)^2 = 9.81 / 0.18$ $T = 0.85 \text{ s}$	A1
3(c)	energy $\propto x_0^2$	C1
	(after 3 cycles,) amplitude = $(0.94)^3 x_0$ $= 0.83x_0$	C1
	ratio final energy / initial energy = 0.83^2 $= 0.69$	A1